AI-Based Lifestyle Intelligence Platform

Jagdish Butte - 96342

Mahim Jain - 96687

Shaurya Pandey - 96728

The Problem

Everyone is tracking. Nobody is connecting the dots.

Sleep Issues

People sleep poorly but don't see how it affects their week

Spending Patterns

People overspend but can't pinpoint why

Habit Cycles

People start habits and quit — with no understanding of the pattern

Existing Tools

- Fitness apps only track steps
- Expense trackers show spending, no insight
- Habit apps are just checklists
- Journaling apps are isolated

The Solution

One platform. Three pillars. AI that connects them.

01

Collect

Daily logs, expenses, journals, mood in one place

02

Connect

Sleep vs. productivity, spending vs. stress

03

Explain

Not just charts, but what the data actually means

04

Advise

Personalized AI recommendations based on your patterns

Example: User asks "Why am I feeling tired?" AI replies: "Your sleep dropped 15% this week. Your water intake averaged 1.8L – below your 2.5L target. Both are impacting your energy levels."

System Architecture

Modular Backend + Polyglot Persistence + External AI Service

System Architecture

Vector DB

Embeddings and search

React Frontend

User interface and client

MongoDB

Logs and journals storage

REST API

HTTP layer connecting frontend

MySQL

Users and expenses data

Spring Boot Backend

Modular monolith services

AI Design: Three-Phase System

Phase 1: Rule-Based Engine

Backend aggregates user data weekly. Simple threshold rules fire deterministic insights. No AI service needed — always on, zero latency.

- avgSleep < 6 hrs → "Insufficient sleep"
- weeklySpending > budget → "Overspending"
- habitCompletion < 50% → "Low consistency"

Phase 2: LLM Insights

Backend sends aggregated summary to FastAPI. LLM generates personalized, context-aware insights. Example: "Your sleep has been declining since Tuesday — your late-night sessions are showing up in your mood data."

Phase 3: RAG Chatbot

User asks natural language question. Backend sends user summary + question to AI service. AI retrieves relevant context → LLM generates response.

Use Case Diagram

Actors and Interactions

Admin has read-only access to system-level data

UI Design Overview

Authentication

Login and register pages with JWT token authentication

Dashboard

Daily intelligence hub with summary cards, charts, and AI insights

Daily Log

Single-page form for activity, meals, habits, and mood tracking

Analytics

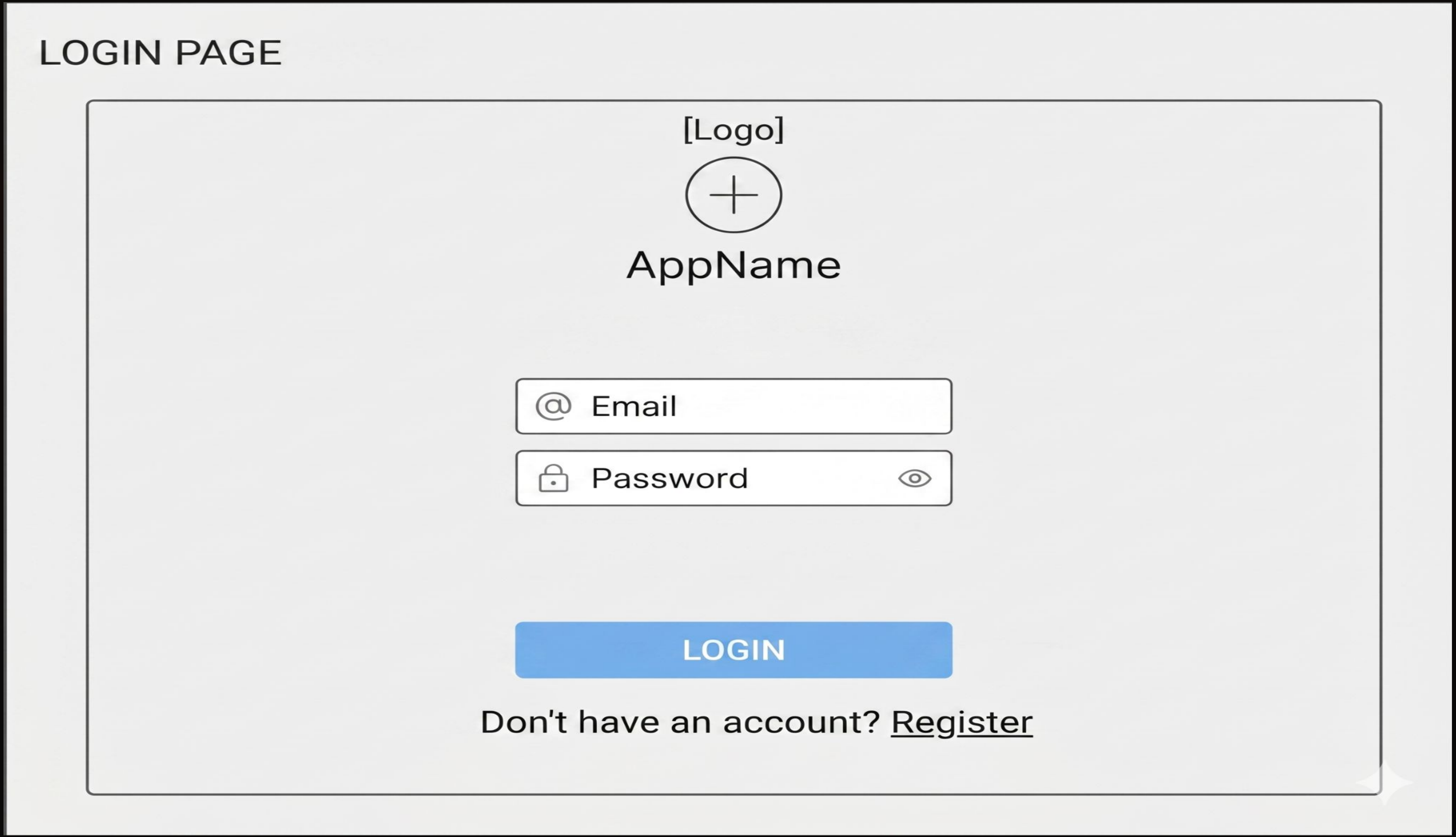
Weekly and monthly trends with date range filters

Expenses

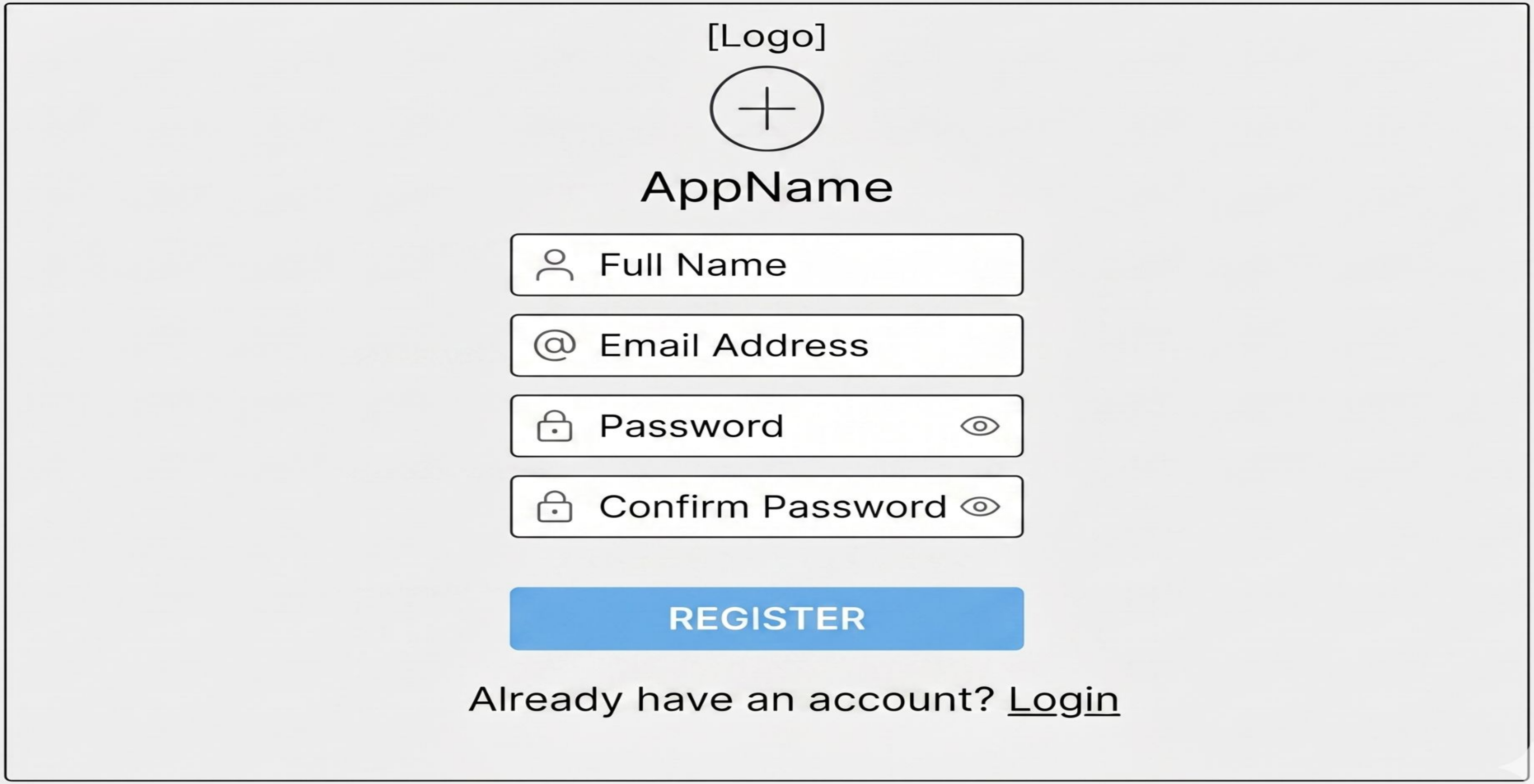
Add, edit, delete expenses with category breakdown

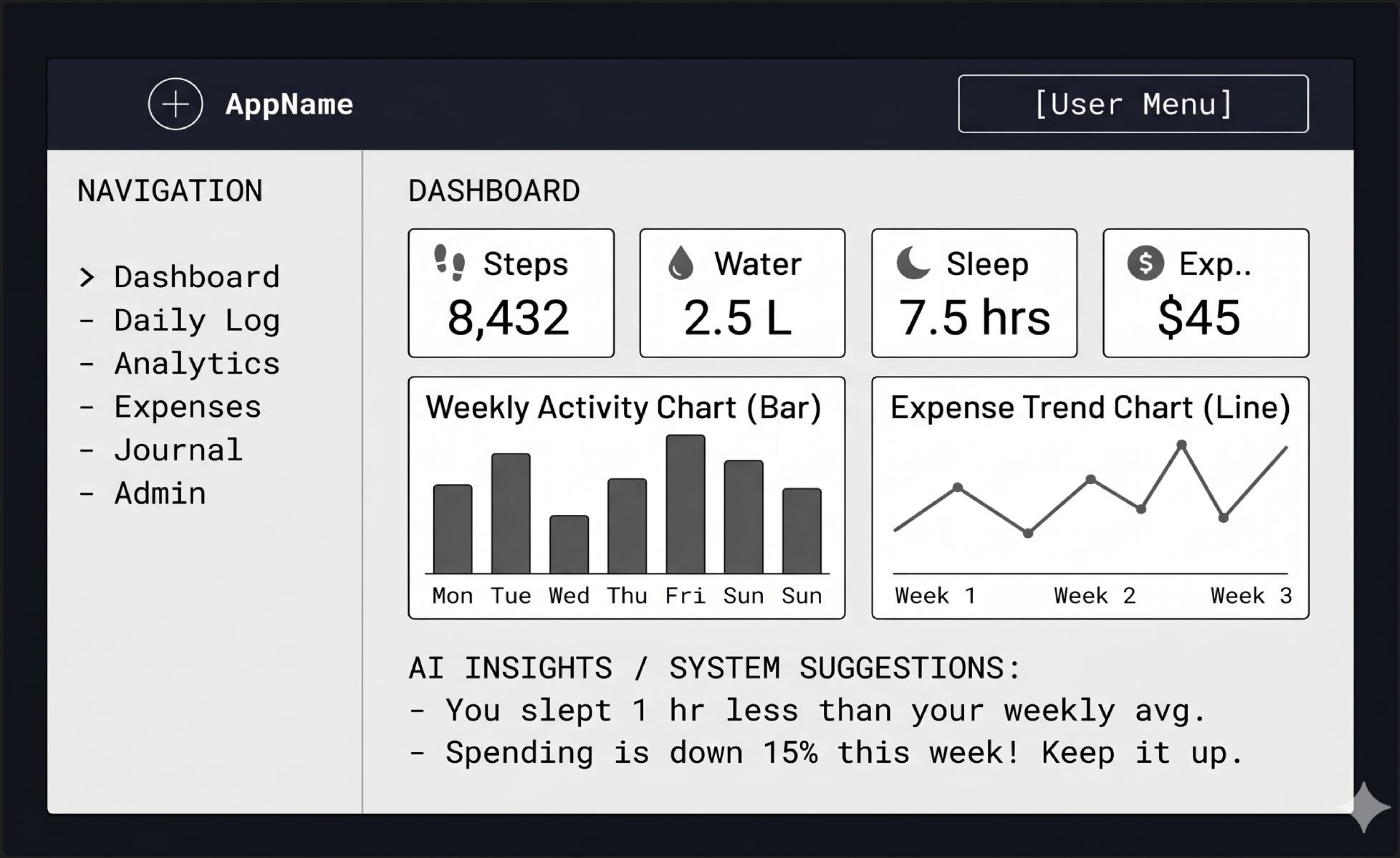
Journal

Record daily thoughts and mood for AI context

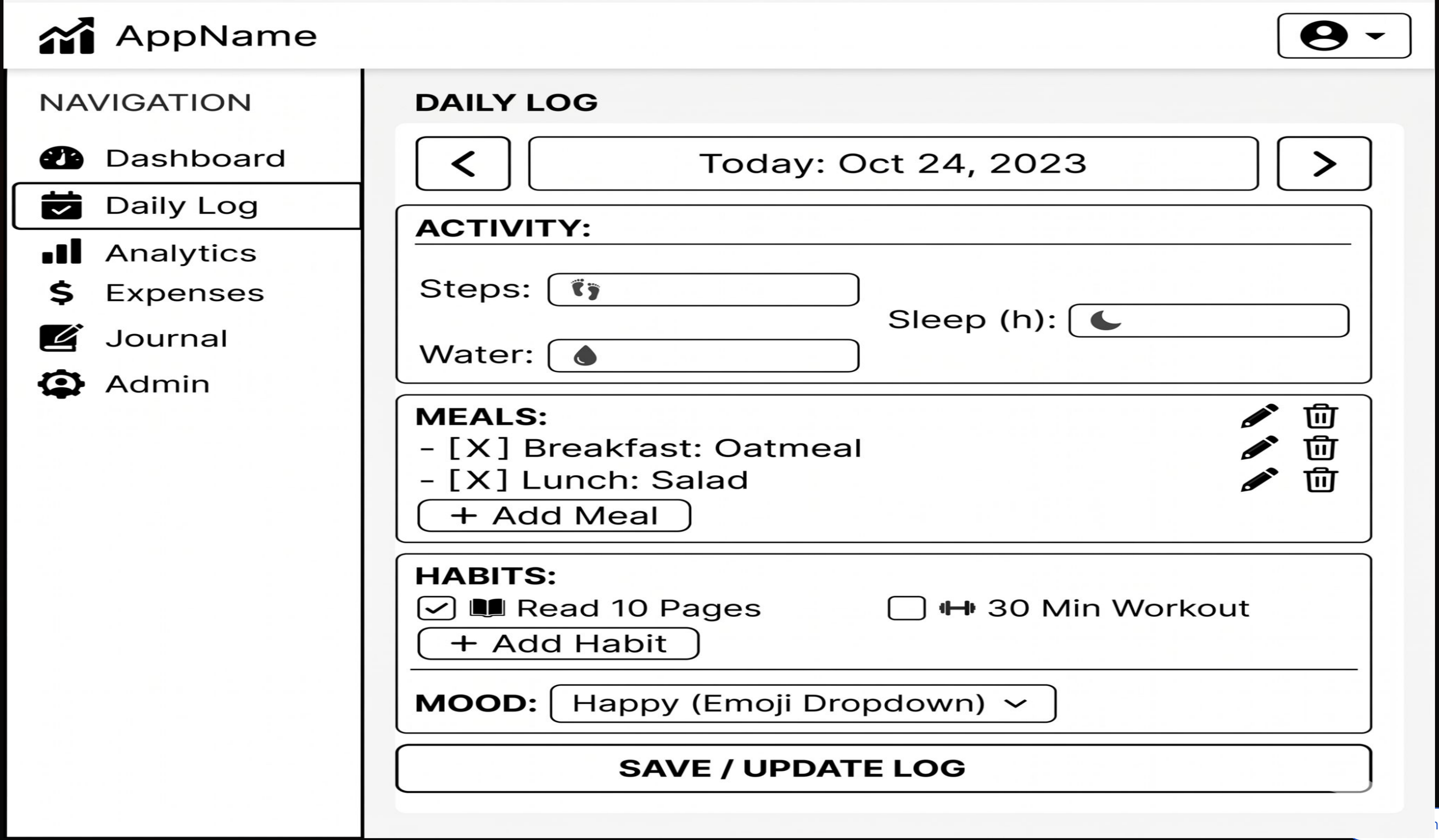

LOGIN PAGE
[Logo]
AppName
Email
Password
LOGIN
Don't have an account? Register

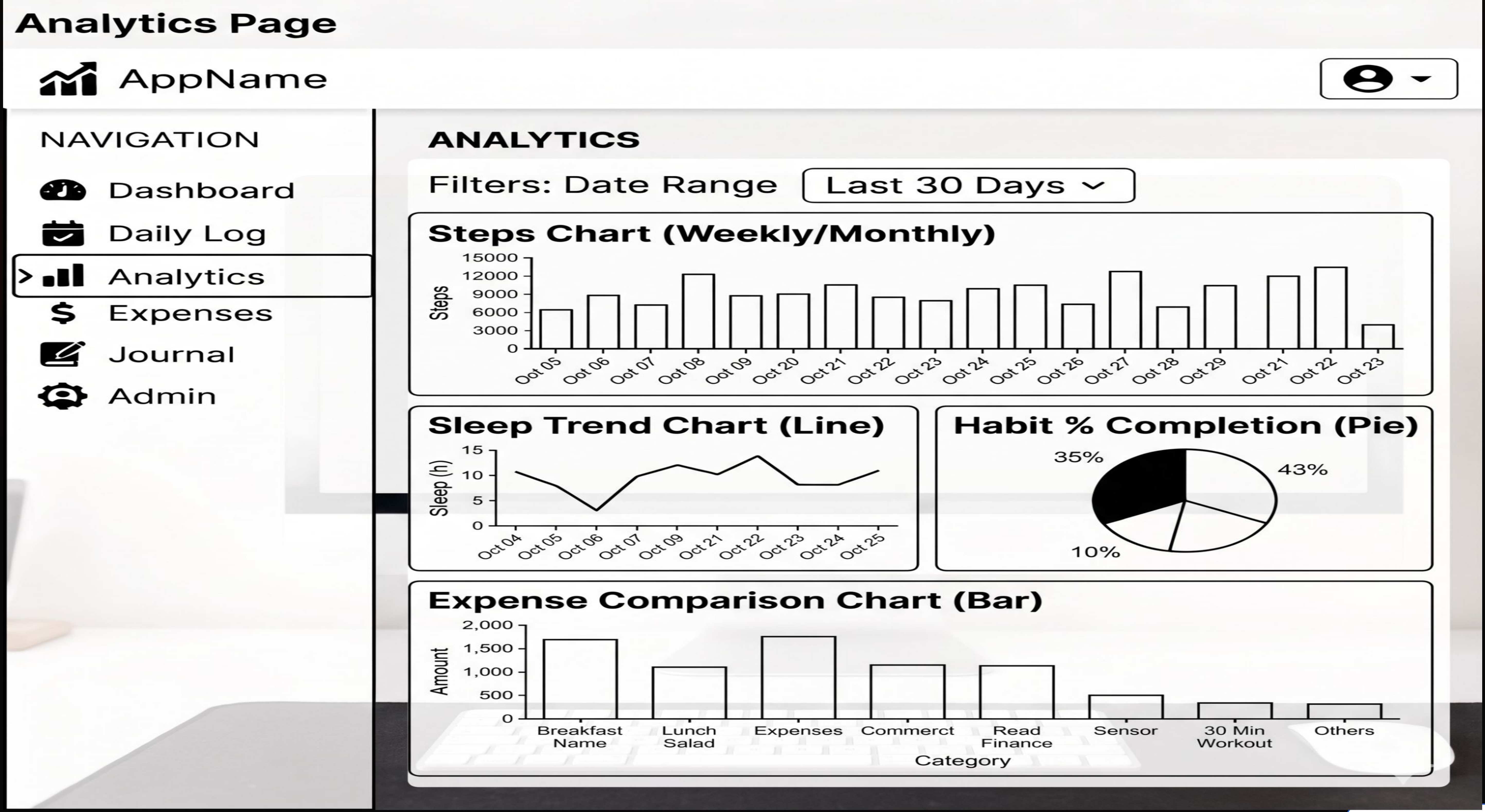
REGISTER PAGE




AppName
[User Menu]
NAVIGATION
> Dashboard
- Daily Log
- Analytics
- Expenses
- Journal
- Admin
DASHBOARD
Steps
8,432
Water
2.5 L
Sleep
7.5 hrs
Exp..
$45
Weekly Activity Chart (Bar)
Mon Tue Wed Thu Fri Sun Sun
Expense Trend Chart (Line)
Week 1 Week 2 Week 3
AI INSIGHTS / SYSTEM SUGGESTIONS:
- You slept 1 hr less than your weekly avg.
- Spending is down 15% this week! Keep it up.

Daily Log Page

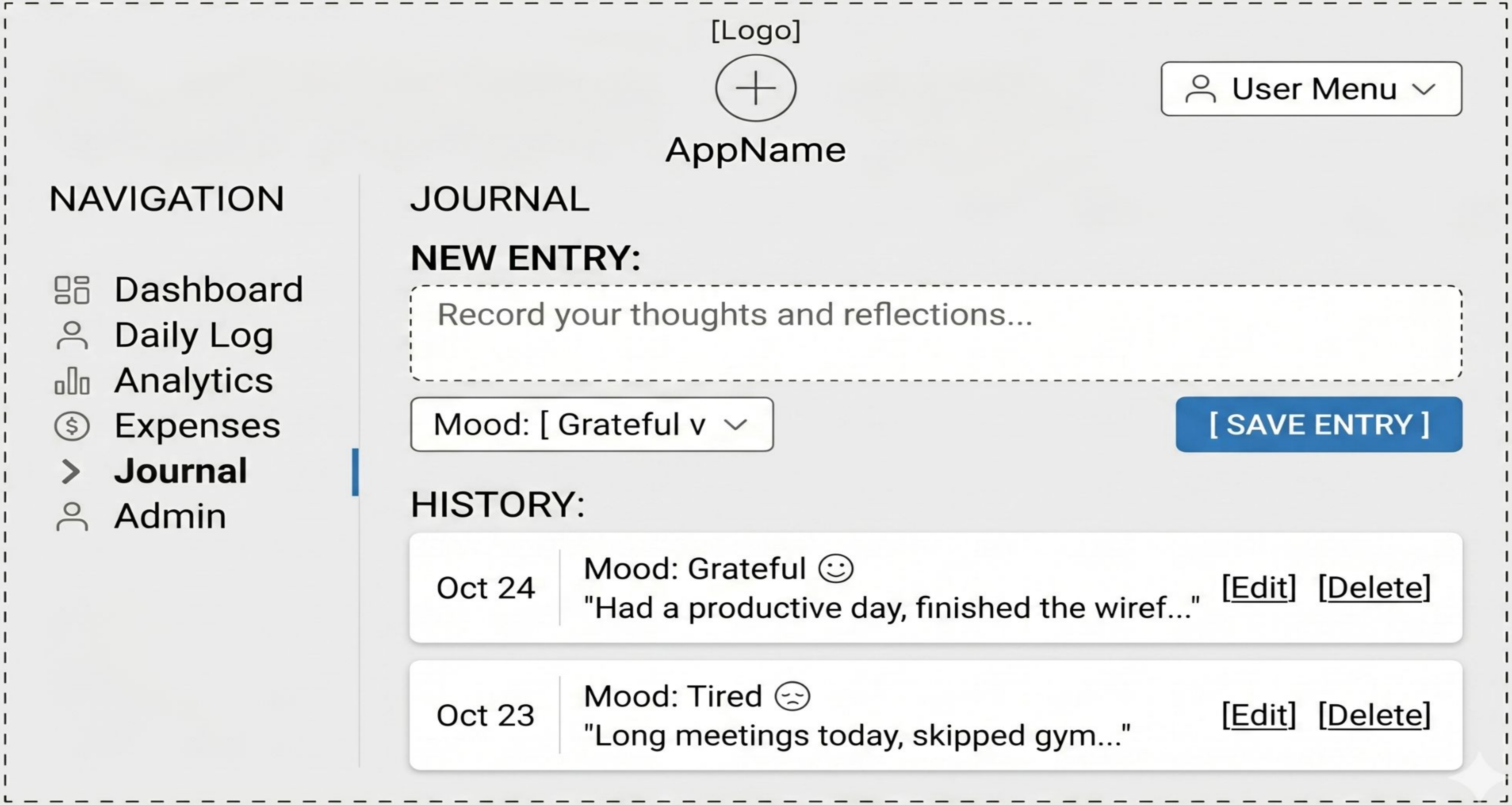




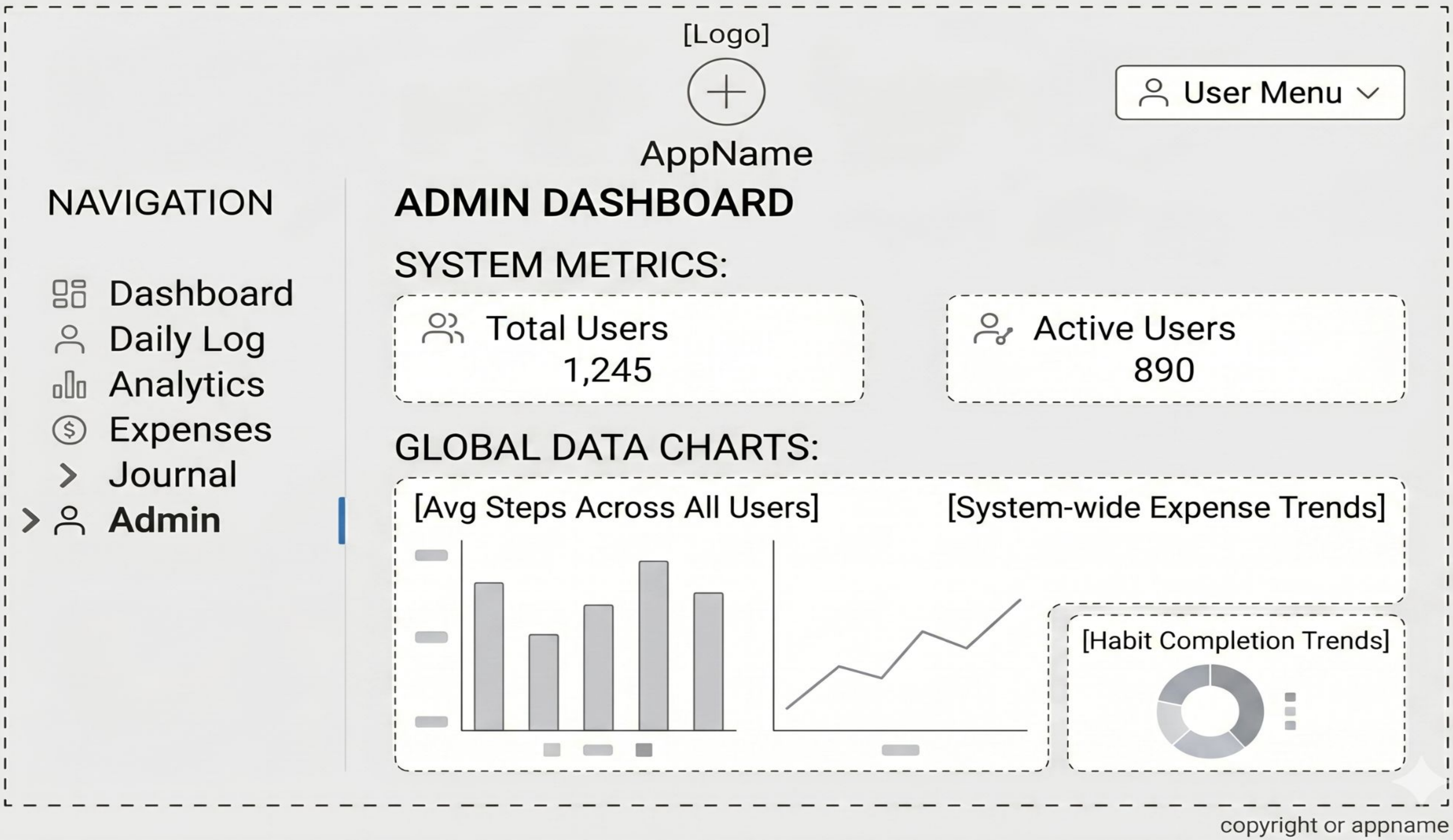
Analytics Page
AppName
NAVIGATION
Dashboard
Daily Log
Analytics
Expenses
Journal
Admin
ANALYTICS
Filters: Date Range
Last 30 Days
Steps Chart (Weekly/Monthly)
Steps
15000
12000
9000
6000
3000
0
Oct 05
Oct 06
Oct 07
Oct 08
Oct 09
Oct 20
Oct 21
Oct 22
Oct 23
Oct 24
Oct 25
Oct 26
Oct 27
Oct 28
Oct 29
Oct 21
Oct 22
Oct 23
Sleep Trend Chart (Line)
Sleep (h)
15
10
5
0
Oct 04
Oct 05
Oct 06
Oct 07
Oct 09
Oct 21
Oct 22
Oct 23
Oct 24
Oct 25
Habit % Completion (Pie)
35%
43%
10%
Expense Comparison Chart (Bar)
Amount
2,000
1,500
1,000
500
0
Breakfast Name
Lunch Salad
Expenses
Commerct
Read Finance
Sensor
30 Min Workout
Others
Category

EXPENSES PAGE

JOURNAL PAGE



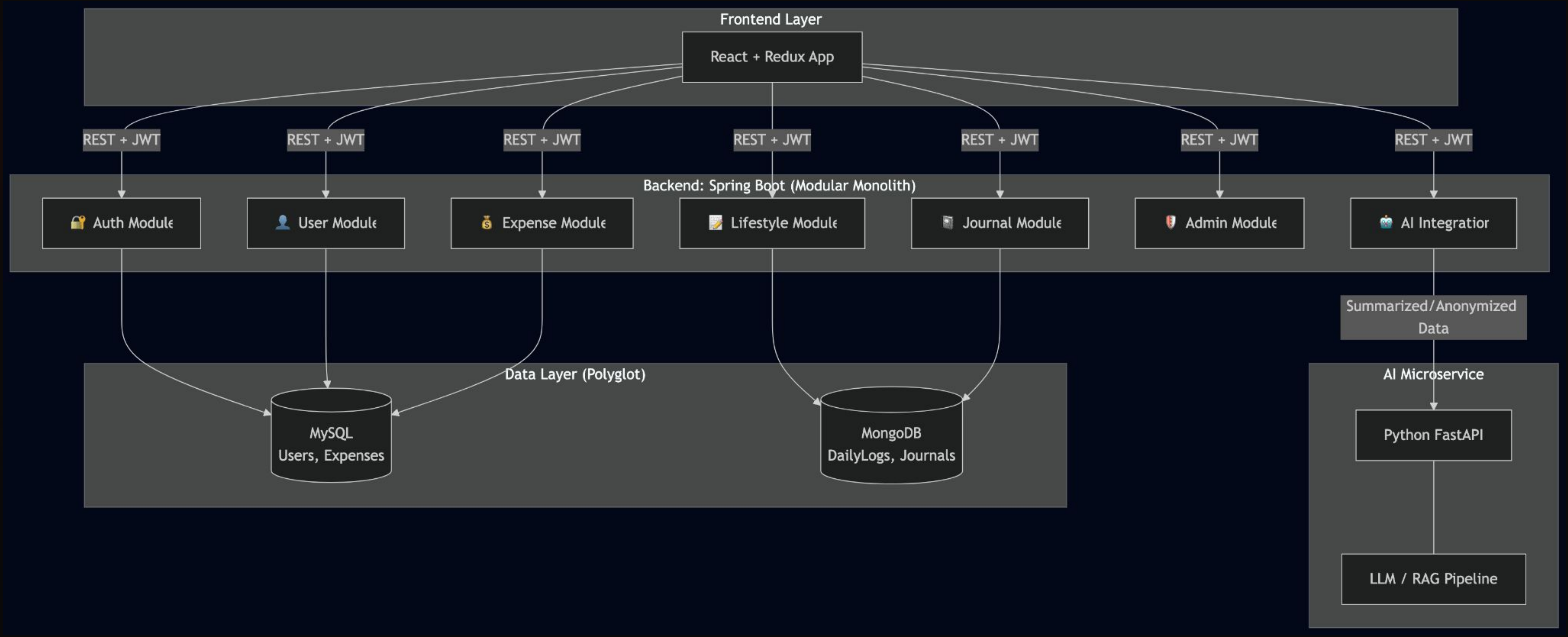
ADMIN PAGE


[Logo]
AppName
User Menu
NAVIGATION
Dashboard
Daily Log
Analytics
Expenses
Journal
Admin
ADMIN DASHBOARD
SYSTEM METRICS:
Total Users
1,245
Active Users
890
GLOBAL DATA CHARTS:
[Avg Steps Across All Users]
[System-wide Expense Trends]
[Habit Completion Trends]
copyright or appname

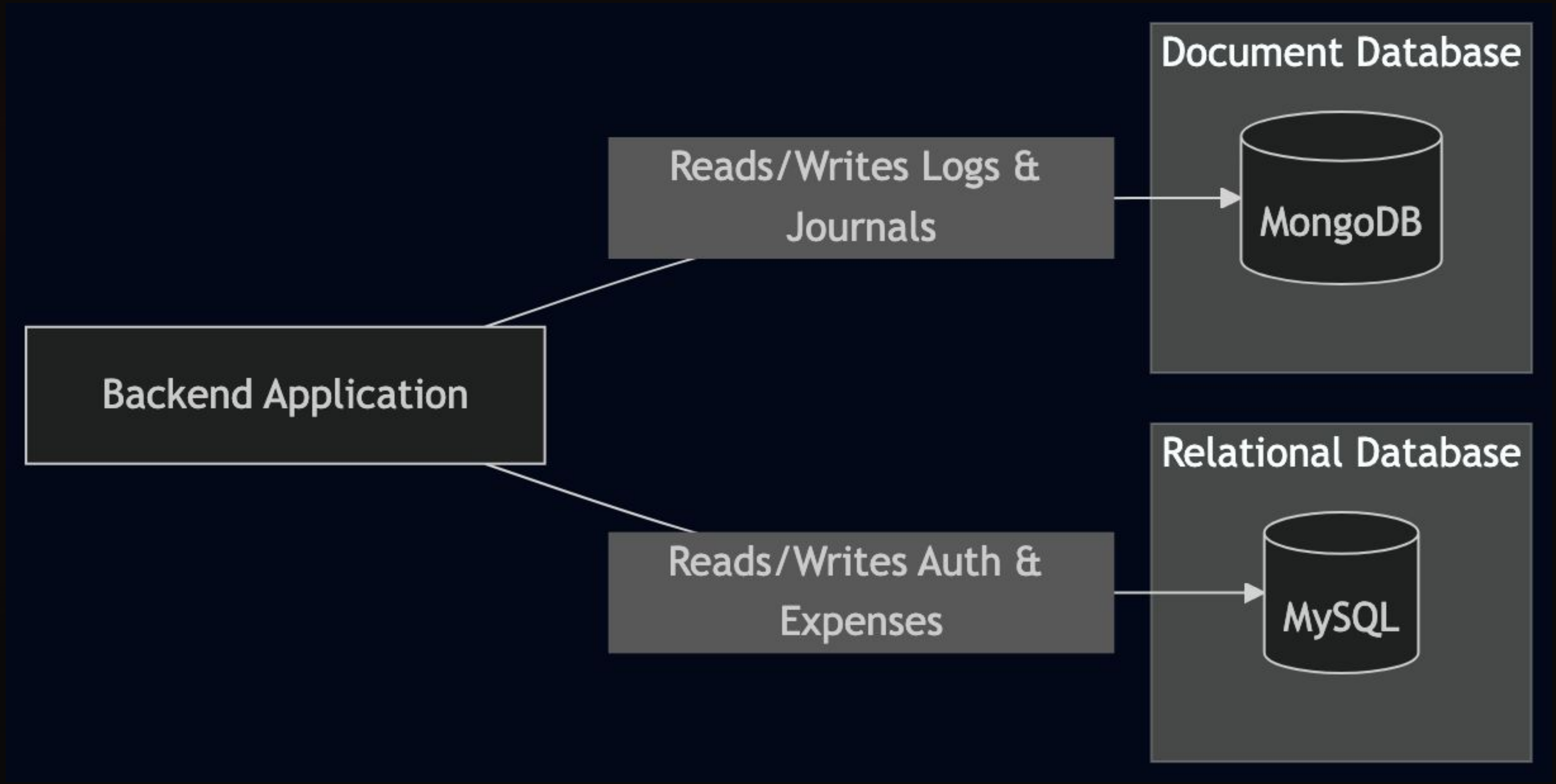
Project's Tech Stack

Layer	Technology	Why
Frontend	React + Redux	Component-based, state management
Backend	Java + Spring Boot	Modular monolith architecture
Relational DB	MySQL	Structured data, ACID compliance
NoSQL DB	MongoDB	Flexible schema for lifestyle data
AI Service	Python FastAPI	LLM ecosystem, lightweight
Auth	JWT	Stateless, scalable token auth
Deployment	Docker + Vercel + AWS	Container isolation, easy hosting

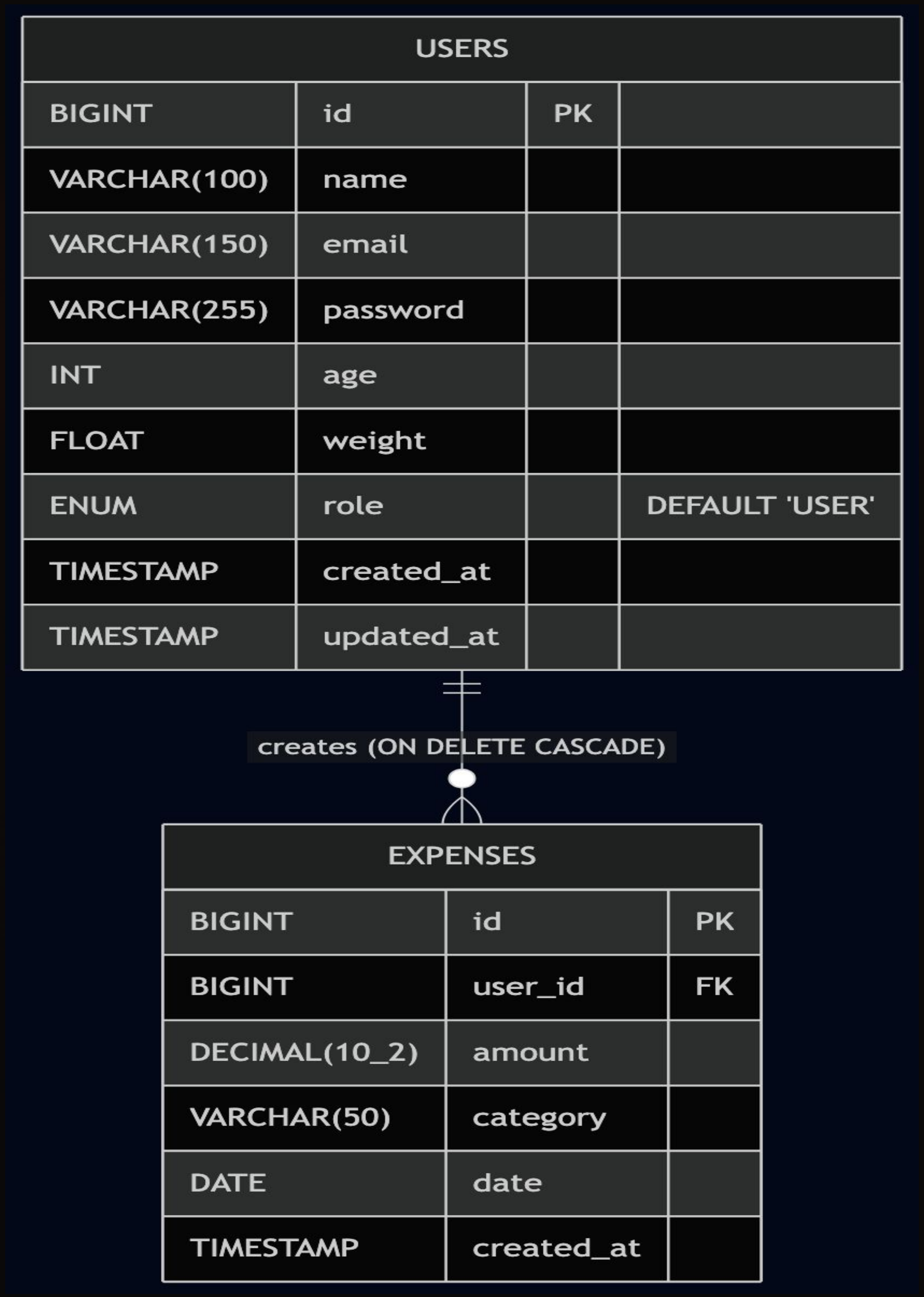
Backend Architecture & Data Flow

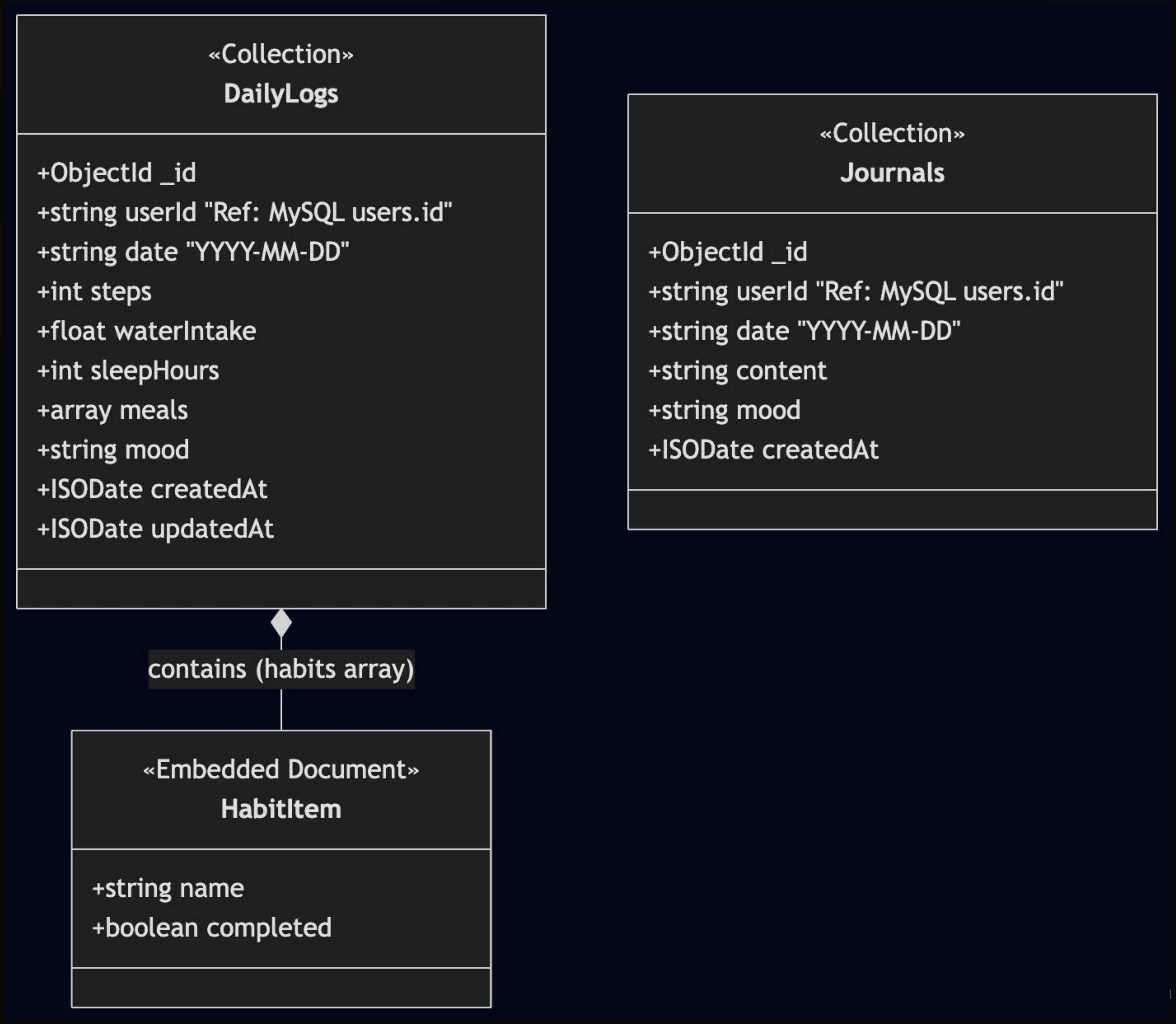


Database Design



Database Schema





REST API – 8 Module Groups, JWT Protected

Module	Key Endpoints
Auth	POST /auth/register, POST /auth/login
User	GET /users/profile, PUT /users/profile
Daily Log	POST /daily-log (upsert), GET /daily-log?date=
Expense	POST /expenses, GET /expenses?range=, PUT/DELETE /expenses/{id}
Journal	POST /journals, GET /journals, PUT/DELETE /journals/{id}
Dashboard	GET /dashboard, GET /analytics?range=weekly
AI	GET /ai/insights, POST /ai/chat
Admin	GET /admin/stats

End-to-End Data Flow – How It All Connects

STEP 01

User Authentication

User logs in → JWT token issued for secure session management.

STEP 02

Daily Log Input

User logs daily data (steps, sleep, water, meals, habits) → Stored in MongoDB (DailyLogs).

STEP 03

Financial Tracking

User adds expense → Stored in MySQL (Expenses) via relational mapping.

STEP 04

Dashboard Aggregation

Dashboard loads → Backend aggregates MongoDB + MySQL data → Frontend renders charts + summary cards.

STEP 05

AI Insights Generation

Backend computes weekly averages → Sends summary to AI Service (FastAPI) → LLM generate insights.

STEP 06

Interactive AI Chat

User chats → Context sent to AI Service → RAG pipeline retrieves + generates response → Shown in panel.

Strategic Roadmap: The Path Forward

PHASE 02

Database Implementation

Development of SQL scripts and NoSQL collection structures for high-performance data storage.

PHASE 03

Spring Boot Backend Modules

Building the core REST API services, security protocols, and business logic integration.

PHASE 04

React Frontend Development

Designing and coding all pages with a focus on responsive UI and interactive data visualization.

PHASE 05

AI Service & LLM Integration

FastAPI deployment with RAG pipeline for personalized lifestyle insights and chat support.

PHASE 06

Testing + Deployment

Comprehensive QA, security auditing, and cloud-native deployment for public release.

THANK YOU

The Path Forward is Powered by Intelligence!